CAMBRIDGE IGCSE CHEMISTRY 0620

Topic 10: Chemistry of the environment

Premium Answer Booklet

20 questions	2020-2025 Paper 4	Premium readable layout

Rebuilt for easier reading with structured tables, clear answer sections and highlighted workings. Use beside the matching topical question bank.

Contents

Question 01 | 17 marks

2020 May/June Paper 41 Question 3

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 7: Acids, bases and salts, Topic 6: Chemical reactions, Topic 9: Metals
Answer source label: 0620_s20_ms_41.pdf

3(a)	• zinc blende
3(b)(i)	• reaction is reversible • rate of forward reaction = rate of reverse reaction
3(b)(ii)	• vanadium(V) oxide
3(b)(iii)	• increases the rate of reaction
3(b)(iv)	• particles have more energy (E) • rate of collisions increase • a higher proportion of particles have energy greater than activation energy (E > EA)
3(b)(v)	• exothermic
3(b)(vi)	• Mr of SO3 = 80 • 40% • x • =
3(c)	• anhydrous copper(II) sulfate • carbon
3(d)(i)	• not all (C-C) bonds are single
3(d)(ii)	• C3H7OH -> C3H6 + H2O • C3H6 • rest of the equation
3(d)(iii)	• propene

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
• escape.

END OF ANSWER

Question 02 — 16 marks

2020 May/June Paper 42 Question 3

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_s20_ms_42.pdf

3(a)(i)	• heat in air
3(a)(ii)	• 2SO2 + O2 -> 2SO3 (1) • 450°C (1) • vanadium(V) oxide (1)
3(a)(iii)	• rate of forward reaction and rate of backward reaction are equal (1) • concentrations of reactants and products are constant (1)
3(b)	• concentrated sulfuric acid
3(c)	• H2S2O7 + H2O ->2H2SO4
3(d)	• Cu + 2H2SO4 ->CuSO4 + SO2 + 2H2O
3(e)	• purple to colourless
3(f)	• 2NH3 + H2SO4 ->(NH4)2SO4
3(g)(i)	• barium nitrate / barium chloride
3(g)(ii)	• Ba2+(aq) + SO42-(aq) -> BaSO4(s) • formulae (1) state symbols (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
• escape.

END OF ANSWER

Question 03 — 18 marks

2020 May/June Paper 43 Question 2

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_s20_ms_43.pdf

2(a)(i)	• reversible reaction
2(a)(ii)	• hydrocarbons (reacting with steam)
2(b)	• [increasing pressure] increases yield • [increasing temperature] decreases yield
2(c)	• (particles) have more energy OR (particles) move faster • more collisions per second OR greater collision rate • more (of the) particles OR collisions have sufficient energy / activation energy to react OR a greater percentage / • proportion / fraction of collisions are successful
2(d)(i)	• N / NH_3 • change in oxidation state of N from -3 to +2 / increase in oxidation number / gain in oxygen / loss of electrons
2(d)(ii)	• No separate point listed.
2(d)(iii)	• (it could react with rain water to) form nitric acid / acid rain
2(d)(iv)	• (Mr of HNO_3 =) 63 (1) • 20 (1) • 5 (1) • 120 (dm^3) (1)
2(e)	• horizontal product energy line at lower energy level than reactant • label of product • correct direction of vertical arrow - arrow must start level with reactant energy and finish level with product level • and one arrow head ONLY

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.

END OF ANSWER

Question 04

8 marks

2020 May/June Paper 43 Question 4

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 12: Experimental techniques and chemical analysis
Answer source label: 0620_s20_ms_43.pdf

4(a)	• (filtration:) remove solids from water / remove insoluble substances • (chlorination:) sterilises / kill microbes / prevent illness
4(b)(i)	• white to blue
4(b)(ii)	• higher boiling point / greater than 100°C
4(c)(i)	• more than one spot
4(c)(ii)	• 0.8 (circled)
4(c)(iii)	• use a locating agent

Easy explanation / reminder
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
• Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
• reduced by carbon/CO.

END OF ANSWER

Question 05 — 10 marks

2020 Oct/Nov Paper 41 Question 4

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 2: Atoms, elements and compounds, Topic 7: Acids, bases and salts
Answer source label: 0620_w20_ms_41.pdf

4(a)	• No separate point listed.
4(b)(i)	• fractional (1) • distillation (1)
4(b)(ii)	• (different) boiling point
4(c)	• 2 double bonds (1) • whole molecule correct (2 pairs of lone pairs on each O) (1)
4(d)	• increase in (concentrations of) carbon dioxide • (carbon dioxide is) greenhouse gas/greenhouse effect • contributes to climate change/global warming
4(e)	• photosynthesis

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 06 | 20 marks

2020 Oct/Nov Paper 43 Question 3

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w20_ms_43.pdf

3(a)	• triple bond (1) • diagram completely correct (1)
3(b)(i)	• METHOD 1 • liquid air (1) • fractional distillation (1) • METHOD 2 • hydrogen • burns in air (to remove the oxygen and then scrub out the carbon dioxide)
3(b)(ii)	• (pressure) 200 atmospheres (1) • (temperature) 450 °C (1) • iron catalyst (1) • N2 + 3H2 -> 2NH3 (1) • equilibrium / reversible (1)
3(c)(i)	• substance that speeds up a reaction / increases rate (1) • unchanged (chemically) at the end • OR not used up • OR lowers activation energy (1)
3(c)(ii)	• gain of oxygen / loss of hydrogen / electron loss / increase in oxidation state (oxidation number)
3(c)(iii)	• effect on the rate of the • forward reaction • effect on the equilibrium • yield of NO(g) • increase (1) • decrease (1) • increase (1) • decrease (1)
3(d)	• 4NO2 + O2 + 2H2O -> 4HNO3 • all formulae (1) • equation fully correct(1)
3(e)	• (Mr of NH4NO3 =) 80 (1) • 35% (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
• escape.

END OF ANSWER

Question 07 | 12 marks

2021 Oct/Nov Paper 41 Question 4

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 6: Chemical reactions
Answer source label: 0620_w21_ms_41.pdf

4(a)	• (4FeCuS 13O2) -> 2Fe2O3 + 4CuO + 8SO2 • Fe2O3 and CuO as a product (1) • Equation fully correct (1)
4(b)(i)	• rate of forward reaction = rate of reverse reaction (1) • concentration of reactants and products are constant (1)
4(b)(ii)	• 450 °C (1) • 1-2 atm (1)
4(b)(iii)	• vanadium(V) oxide
4(b)(iv)	• increase rate of reaction
4(b)(v)	• equilibrium shifts to left hand side (1) • forward reaction is exothermic (1)
4(c)	• C6H12O6 -> 6C + 6H2O • H2O (1) • balance (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
• escape.

END OF ANSWER

Question 08

9 marks

2021 Oct/Nov Paper 43 Question 1

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 11: Organic chemistry, Topic 9: Metals
Answer source label: 0620_w21_ms_43.pdf

1(a)	• oxygen
1(b)	• bauxite
1(c)	• sulfur dioxide
1(d)	• ethanol
1(e)	• sodium chloride
1(f)	• ammonia
1(g)	• carbon dioxide
1(h)	• carbon monoxide
1(i)	• ammonia

Easy explanation / reminder
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
• Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO2 and H2O when oxygen is in excess.

END OF ANSWER

Question 09

10 marks

2022 May/June Paper 41 Question 4

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_s22_ms_41.pdf

4(a)(i)	• $S + O_2 \rightarrow SO_2$
4(a)(ii)	• (temperature) 450 °C (1) • (pressure) 1-2 atmosphere(s) (1) • vanadium(V) oxide catalyst (1) • $2SO_2 + O_2$ ■ $2SO_3$ (1)
4(a)(iii)	• $SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$
4(a)(iv)	• water
4(b)(i)	• 2 bonding pairs as one dot and cross each (1) • 2 lone pairs on S (and no additional electrons on Hs) to complete the outer shell on S and both Hs (1)
4(b)(ii)	• $2H_2S + SO_2 \rightarrow 3S + 2H_2O$

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.

END OF ANSWER

Question 10

16 marks

2022 Oct/Nov Paper 42 Question 3

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 9: Metals, Topic 7: Acids, bases and salts, Topic 6: Chemical reactions
Answer source label: 0620_w22_ms_42.pdf

3(a)	• zinc blende
3(b)	• strong heating in air / roasting in air
3(c)	• contact
3(d)(i)	• 450 (°C) (1) • 1-2 (atm) (1)
3(d)(ii)	• vanadium(V) oxide
3(d)(iii)	• the rate of forward reaction equals (the rate of the) reverse reaction (1) • concentrations of reactants and products are constant (1)
3(d)(iv)	• increased temperature: • (position of) equilibrium moves to left-hand side (1) • reaction is exothermic (1) • increased pressure: • (position of) equilibrium moves to right-hand side (1) • more (gaseous) moles / molecules on left-hand side (1)
3(d)(v)	• rate increases and particles have more energy (1) • more collisions (between particles) occur per second / per unit time (1) • more of the particles / collisions have energy equal to or above the activation energy (1) • or • more of the particles / collisions have sufficient energy to react • or • a higher percentage / proportion / fraction of collisions (of particles) are successful / have energy equal to or above • activation energy
3(e)	• ammonium sulfate

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
• escape.

END OF ANSWER

Question 11

17 marks

2022 Oct/Nov Paper 43 Question 5

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 7: Acids, bases and salts, Topic 3: Stoichiometry
Answer source label: 0620_w22_ms_43.pdf

5(a)(i)	• 450 (1) • 1-2 (1) • vanadium (V) oxide (1)
5(a)(ii)	• SO3 + H2SO4 -> H2S2O7
5(a)(iii)	• water
5(b)	• carbon
5(c)(i)	• 2.5 -> 10-3 / 0.0025 (1) • 1.25 -> 10-3 / 0.00125 (1) • 20 (1)
5(c)(ii)	• No separate point listed.
5(d)(i)	• lilac flame
5(d)(ii)	• Any Two from • solid disappears • blue solution • bubbles / effervescence / fizzing
5(d)(iii)	• white precipitate
5(e)	• Ba2+(aq) + SO42-(aq) -> BaSO4(s) • BaSO4 on the right (1) • ONLY Ba2+ and SO42- on the left (1) • state symbols (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.

END OF ANSWER

Question 12

11 marks

2023 Feb/March Paper 42 Question 1

Main topic: Topic 10: Chemistry of the environment
Answer source label: 0620_m23_ms_42.pdf

1(a)	• carbon dioxide
1(b)	• 78 (%)
1(c)	• calcium oxide
1(d)	• catalytic converter
1(e)(i)	• CH4
1(e)(ii)	• lowest relative molecular mass
1(f)	• toxic
1(g)	• glucose • oxygen
1(h)	• two dot-and-cross double bonds • two pairs of non-bonding electrons on O and zero non-bonding electrons on C

Easy explanation / reminder
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 13

18 marks

2023 Feb/March Paper 42 Question 3

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 6: Chemical reactions, Topic 5: Chemical energetics
Answer source label: 0620_m23_ms_42.pdf

3(a)	• nitrogen: air (1) • hydrogen: methane (1)
3(b)(i)	• enthalpy change
3(b)(ii)	• (the value of) ΔH is negative
3(b)(iii)	• 450 (1) • 20 000 (1) • iron (1)
3(b)(iv)	• one mark for each of • decreases • decreases • decreases • no change
3(b)(v)	• kinetic energy of particles increases (1) • frequency of collisions between particles increases (1) • higher percentage / proportion / fraction of collisions / particles have energy greater than / equal to activation energy (1) • or • more of the collisions / particles have energy greater than / equal to activation energy
3(c)(i)	• H2SO4
3(c)(ii)	• fertiliser • February/March 2023
3(c)(iii)	• Mr of (NH4)2SO4 = 132 (1) • 2 -> 14 = 28 and %N = 100 -> 28 / 132 = 21.2% (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
• escape.

END OF ANSWER

Question 14

15 marks

2023 May/June Paper 43 Question 6

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_s23_ms_43.pdf

6(a)(i)	• air
6(a)(ii)	• methane
6(a)(iii)	• 450 (°C) • 200 (atm)
6(a)(iv)	• iron
6(a)(v)	• (a substance which) increases the rate of a reaction • remains unchanged at the end of the reaction
6(b)(i)	• temperature change: • low(er) rate (of reaction) • pressure change: • (position of) equilibrium shifts to the left hand side/ towards reactants
6(b)(ii)	• 4NO + 3O2 + 2H2O ->.4HNO3
6(c)(i)	• CuCO3 + 2HNO3 -> Cu(NO3)2 + CO2 + H2O • Cu(NO3)2 • correct equation
6(c)(ii)	• undissolved solid • effervescence stops on addition of more copper(II) carbonate
6(d)(iii)	• copper(II) oxide or • copper(II) hydroxide

Easy explanation / reminder
• Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
• escape.
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.

END OF ANSWER

Question 15

6 marks

2023 Oct/Nov Paper 41 Question 1

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 11: Organic chemistry, Topic 7: Acids, bases and salts
Answer source label: 0620_w23_ms_41.pdf

1(a)	• sulfur dioxide
1(b)	• ammonia
1(c)	• xenon
1(d)	• oxygen
1(e)	• ethene
1(f)	• ammonia

Easy explanation / reminder
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
• Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO2 and H2O when oxygen is in excess.

END OF ANSWER

Question 16

21 marks

2023 Oct/Nov Paper 43 Question 3

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 6: Chemical reactions, Topic 2: Atoms, elements and compounds
Answer source label: 0620_w23_ms_43.pdf

3(a)(i)	• No separate point listed.
3(a)(ii)	• natural gas
3(a)(iii)	• three single bonding pairs containing one dot and one cross (1) • two dots or two crosses on N (and no additional electrons on Hs) to complete the outer shell on N and all 3 Hs (1)
3(a)(iv)	• $N_2 + 3H_2 \rightarrow 2NH_3$ (1) • ■ (1) • 450 °C (1) • 200 atmospheres / 20 000 kPa (1) • iron (1)
3(b)(i)	• platinum
3(b)(ii)	• -3 (1) • +2 (1)
3(b)(iii)	• oxidation AND oxidation number increases
3(b)(iv)	• increase (1) • decrease (1) • increase (1) • no change (1)
3(b)(v)	• kinetic energy of particles is lower (1) • frequency of collisions between particles is lower (1) • fewer / lower percentage / lower proportion / lower fraction of particles have energy greater than / equal to activation • energy • OR • fewer / lower percentage / lower fraction of collisions have energy greater than / equal to activation energy (1)
3(c)	• $4NO + 3O_2 + 2H_2O \rightarrow 4HNO_3$

Easy explanation / reminder
• Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
• escape.
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.

END OF ANSWER

Question 17

7 marks

2024 May/June Paper 42 Question 1

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 9: Metals
Answer source label: 0620_s24_ms_42.pdf

1(a)(i)	• methane
1(a)(ii)	• nitrogen dioxide
1(a)(iii)	• propene
1(a)(iv)	• helium
1(a)(v)	• carbon monoxide
1(b)	• 2NO2 + 4CO -> N2 + 4CO2 • N2 + CO2 as only products • correct equation

Easy explanation / reminder
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
• Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
• reduced by carbon/CO.

END OF ANSWER

Question 18

6 marks

2025 Feb/March Paper 42 Question 1

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 2: Atoms, elements and compounds
Answer source label: 0620_m25_ms_42.pdf

1(a)	• No separate point listed.
1(b)	• No separate point listed.
1(c)	• No separate point listed.
1(d)	• No separate point listed.
1(e)	• No separate point listed.
1(f)	• 6.02 -> 1023

Easy explanation / reminder
• Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO2 and H2O when oxygen is in excess.

END OF ANSWER

Question 19

8 marks

2025 May/June Paper 42 Question 1

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 8: The Periodic Table
Answer source label: 0620_s25_ms_42.pdf

1(a)	• 78(%)
1(b)	• No separate point listed.
1(c)	• No separate point listed.
1(d)	• No separate point listed.
1(e)	• any value between 25 to 35 inclusive
1(f)	• No separate point listed.
1(g)	• No separate point listed.
1(h)	• No separate point listed.

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.

END OF ANSWER

Question 20

13 marks

2025 May/June Paper 43 Question 4

Main topic: Topic 10: Chemistry of the environment
Cross-topic links: Topic 3: Stoichiometry, Topic 7: Acids, bases and salts
Answer source label: 0620_s25_ms_43.pdf

4(a)(i)	• 4FeS2 + 11O2-> 8SO2 + 2Fe2O3
4(a)(ii)	• double bond of 2 dots and 2 crosses • 2 non-bonding pairs of dots on one oxygen atom and 2 non-bonding pairs of crosses on the other oxygen atom
4(b)(i)	• (temperature) • 450 and °C • (pressure) • 2 and atm • OR • 200 and kPa • vanadium(V) oxide catalyst
4(b)(ii)	• 2SO2 + O2 ■ 2SO3
4(b)(iii)	• SO3 + H2SO4 -> H2S2O7
4(c)	• zero • +4
4(d)	• mol NaHCO3 = 4.20 / 84 = 0.05(00) • mol CO2 = 0.05(00) • vol CO2 = 0.05 x 24 000 = 1200

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.

END OF ANSWER